

Air Quality and Temperature Monitoring System

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List of Principal Symbols and Acronyms

Symbol	Meaning
ESP32	Espressif 32-bit Wi-Fi and Bluetooth Microcontroller
MQ-135	Air Quality Gas Sensor
DS18B20	Digital Temperature Sensor
AQI	Air Quality Index
CO	Carbon Monoxide
IoT	Internet of Things
GPIO	General Purpose Input Output
ADC	Analog to Digital Converter
Wi-Fi	Wireless Fidelity
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript

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Abstract

Air pollution is one of the major environmental challenges affecting human health. This project presents an IoT-based Air Quality and Temperature Monitoring System using the ESP32 microcontroller, MQ-135 gas sensor, DS18B20 temperature sensor, Firebase Realtime Database, and a web dashboard.

The system continuously monitors air quality and temperature, uploads the measured data to Firebase through Wi-Fi, and displays real-time information on a web dashboard. Whenever the measured air quality exceeds the predefined threshold, an active buzzer generates an audible warning.

The developed system is low-cost, portable, reliable, and suitable for smart homes, laboratories, educational institutions, industries, and environmental monitoring applications.

Chapter 1

Introduction

1.1 Introduction

Air pollution has become one of the most serious environmental problems due to rapid industrialization, urbanization, and increasing vehicular emissions. Harmful gases released into the atmosphere affect both human health and the environment. Continuous monitoring of air quality and temperature is essential for maintaining a healthy living environment.

Recent developments in the Internet of Things (IoT) have made it possible to develop smart environmental monitoring systems capable of collecting, transmitting, and displaying sensor data in real time. Such systems are cost-effective and suitable for remote monitoring.

The proposed project, entitled **Air Quality and Temperature Monitoring System**, uses the ESP32 microcontroller, MQ-135 gas sensor, DS18B20 temperature sensor, Firebase Realtime Database, and a web dashboard to provide continuous monitoring of air quality and temperature.

1.2 Review of Previous Works (Literature Review)

Several researchers have developed IoT-based environmental monitoring systems using different gas sensors and cloud platforms. Traditional monitoring systems are generally expensive, require complex installation, and are limited to specific locations.

Recent studies have focused on low-cost microcontrollers such as ESP32 and Arduino combined with cloud services like Firebase and Blynk for real-time monitoring. These systems improve accessibility, reduce cost, and enable remote monitoring through the Internet.

1.3 Objectives and Motivation

The main objectives of this project are:

- To monitor air quality using the MQ-135 gas sensor.
- To measure ambient temperature using the DS18B20 sensor.
- To calculate Air Quality Index (AQI).
- To upload sensor data to Firebase Realtime Database.
- To display real-time data on a web dashboard.
- To activate an audible buzzer during hazardous conditions.
- To develop a low-cost IoT-based monitoring system.

The motivation behind this work is to provide an affordable and efficient monitoring solution that can be used in homes, laboratories, educational institutions, industries, and public places.

1.4 Scope of Present Work

The proposed system provides continuous monitoring of air quality and temperature using IoT technology. It supports cloud-based data storage and remote monitoring through a web dashboard.

The system can be further extended by integrating additional sensors, mobile applications, AI-based prediction, GPS tracking, and notification services for smart city and industrial applications.

Chapter 2

System Design and Implementation

2.1 System Overview

The proposed Air Quality and Temperature Monitoring System is designed using the ESP32 Development Board as the main controller. The system continuously measures air quality and temperature using the MQ-135 Gas Sensor and DS18B20 Temperature Sensor. The collected data is processed by the ESP32 and transmitted to Firebase Realtime Database through a Wi-Fi connection. A web dashboard displays the real-time sensor values, while an active buzzer generates an alarm whenever the air quality exceeds the predefined threshold.

2.2 Block Diagram

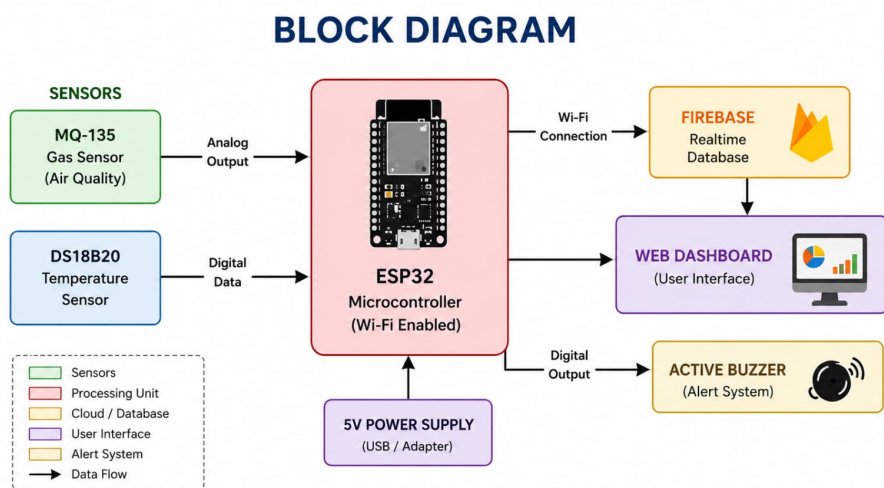


Figure 2.1: Block Diagram of the Proposed System

2.3 Hardware Components

The following hardware components are used in the project.

Sl. No.	Component	Quantity
1	ESP32 DevKit V1	1
2	MQ-135 Gas Sensor	1
3	DS18B20 Temperature Sensor	1
4	Active Buzzer	1
5	Breadboard	1
6	Jumper Wires	As Required
7	4.7k Ω Resistor	1
8	USB Cable	1

Table 2.1: Hardware Components

2.4 Software Requirements

The software used for this project is listed below.

Arduino IDE	Programming ESP32
Firebase Realtime Database	Cloud Storage
HTML/CSS/JavaScript	Web Dashboard
Google Chrome	Dashboard Monitoring
Overleaf	Report Preparation

Table 2.2: Software Requirements

2.5 Circuit Diagram

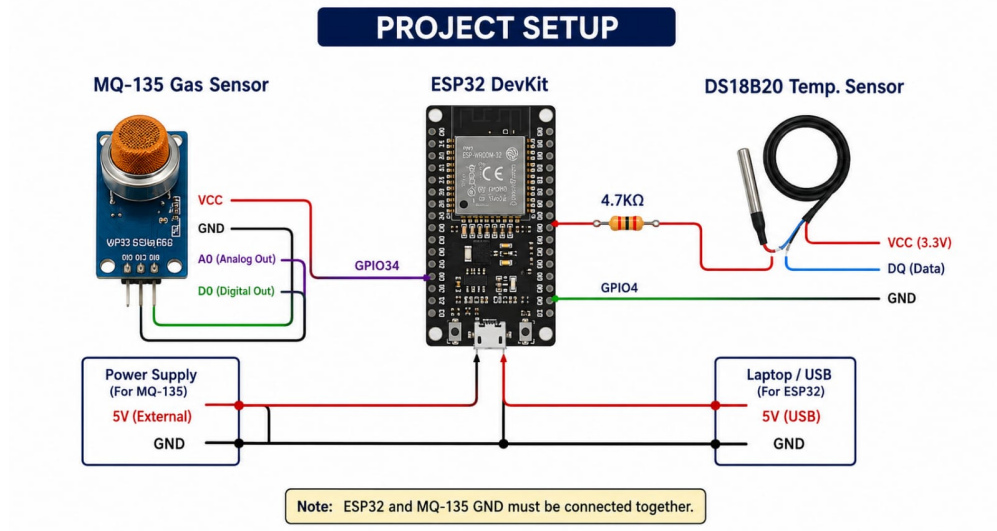


Figure 2.2: Circuit Diagram of the Proposed System

2.6 Pin Configuration

Component	ESP32 Pin
MQ-135 Analog Output	GPIO34
DS18B20 Data Pin	GPIO32
Active Buzzer	GPIO25
MQ-135 VCC	5V
DS18B20 VCC	3.3V
Common Ground	GND

Table 2.3: ESP32 Pin Configuration

2.7 Firebase Realtime Database

Firebase Realtime Database is used as the cloud platform for storing sensor readings. The ESP32 uploads air quality, temperature, AQI, CO level, and buzzer status to Fire-

base through Wi-Fi. The cloud database enables real-time monitoring from any location with internet access.

2.8 Web Dashboard

A web dashboard developed using HTML, CSS, and JavaScript displays the live sensor data received from Firebase. The dashboard shows Carbon Monoxide (CO), Air Quality Index (AQI), Temperature, Emergency Status, and Buzzer Status. This provides a user-friendly interface for monitoring environmental conditions in real time.

Chapter 3

Working Principle

3.1 Introduction

The proposed Air Quality and Temperature Monitoring System continuously monitors the surrounding environment using the MQ-135 Gas Sensor and DS18B20 Temperature Sensor. The ESP32 microcontroller collects the sensor data, processes it, uploads it to Firebase Realtime Database through Wi-Fi, and displays the measured values on a web dashboard. If the measured AQI exceeds the predefined threshold, the active buzzer is automatically activated to warn nearby users.

3.2 Working Principle

The complete operation of the system is carried out in the following steps:

1. Power ON the ESP32 development board.
2. Initialize the MQ-135 Gas Sensor.
3. Initialize the DS18B20 Temperature Sensor.
4. Connect ESP32 to the Wi-Fi network.
5. Read air quality data from MQ-135.
6. Read temperature data from DS18B20.
7. Calculate Carbon Monoxide (CO) concentration.
8. Calculate Air Quality Index (AQI).
9. Compare AQI with the predefined threshold.
10. If AQI exceeds the threshold, turn ON the buzzer.

11. Upload sensor values to Firebase Realtime Database.
12. Display the latest readings on the Web Dashboard.
13. Repeat the monitoring process continuously.

3.3 System Flowchart

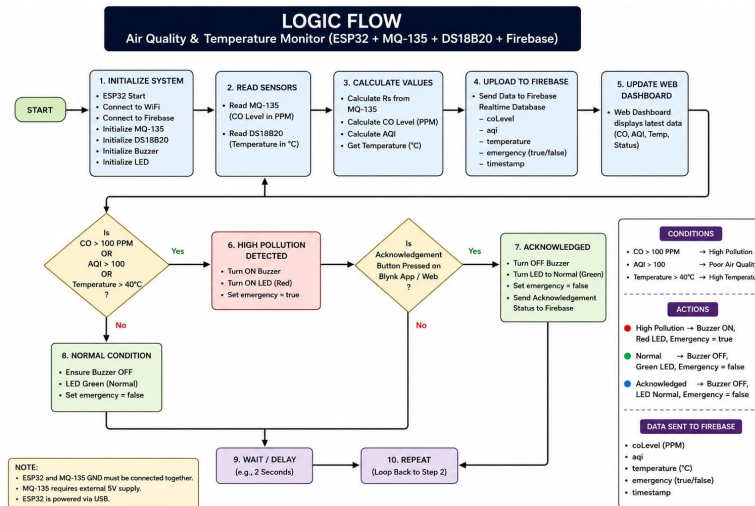


Figure 3.1: Flowchart of the Proposed System

3.4 Algorithm

The algorithm of the proposed system is given below.

1. Start the system.
2. Initialize all sensors.
3. Establish Wi-Fi connection.
4. Read gas sensor value.
5. Read temperature value.
6. Calculate AQI and CO concentration.
7. Update Firebase Database.
8. Refresh Web Dashboard.
9. Check emergency condition.

10. Activate buzzer if necessary.
11. Repeat continuously.

3.5 Data Flow

The MQ-135 sensor continuously measures harmful gases while the DS18B20 measures ambient temperature. The ESP32 processes both sensor values and transmits the information to Firebase Realtime Database via Wi-Fi. The Web Dashboard retrieves the latest values from Firebase and displays them to the user in real time. Whenever the AQI exceeds the threshold value, the buzzer is activated to indicate hazardous air quality.

3.6 Firebase Communication

The ESP32 communicates with Firebase Realtime Database using Wi-Fi. The sensor values are periodically uploaded to the cloud database. The database stores the latest readings and allows multiple users to monitor the system remotely through the web dashboard.

3.7 Web Dashboard Operation

The web dashboard is designed using HTML, CSS, and JavaScript. It provides real-time visualization of:

- Carbon Monoxide (CO)
- Air Quality Index (AQI)
- Temperature
- Buzzer Status
- Emergency Status
- Last Updated Sensor Values

The dashboard automatically refreshes whenever new data is received from Firebase.

Chapter 4

Results and Discussion

4.1 Experimental Setup

The developed prototype was tested under normal indoor environmental conditions. The ESP32 was connected to a Wi-Fi network and interfaced with the MQ-135 Gas Sensor, DS18B20 Temperature Sensor, and an Active Buzzer. Sensor readings were transmitted to Firebase Realtime Database and displayed on the web dashboard in real time.

4.2 Firebase Output

The ESP32 successfully uploaded all sensor data to Firebase Realtime Database. The uploaded parameters include:

- Air Quality Index (AQI)
- Carbon Monoxide (CO)
- Temperature
- Buzzer Status
- Emergency Status

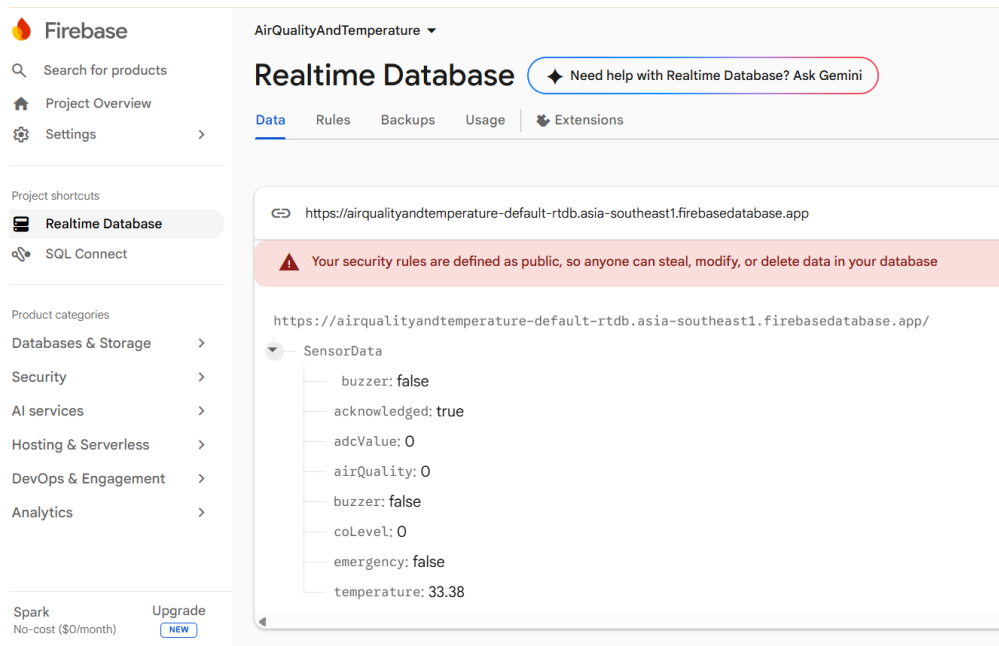


Figure 4.1: Firebase Realtime Database Output

4.3 Web Dashboard Output

The web dashboard displays the latest sensor readings in real time. Users can monitor environmental conditions remotely through the dashboard.

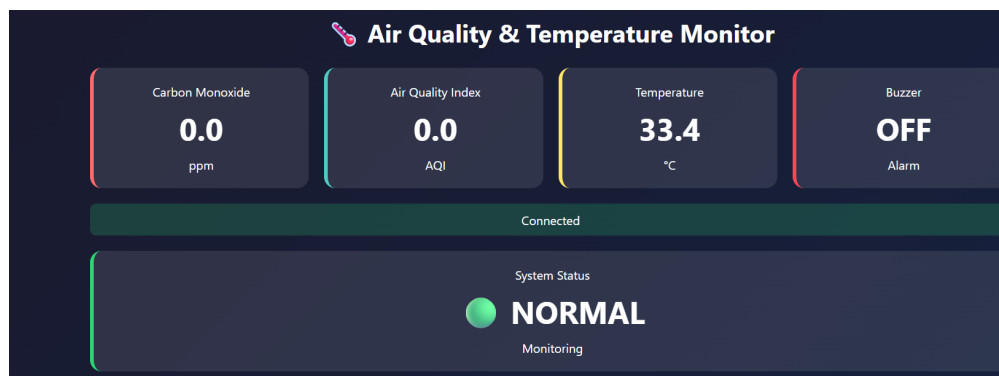


Figure 4.2: Web Dashboard Output

4.4 Observation Table

Reading	AQI	CO (ppm)	Temperature (°C)	Buzzer
1	65	2.4	29.1	OFF
2	78	3.2	29.5	OFF
3	115	5.7	30.2	ON
4	132	6.4	30.8	ON
5	71	2.9	29.3	OFF

Table 4.1: Experimental Observation

4.5 Performance Analysis

The developed system continuously monitored air quality and temperature with satisfactory accuracy. Sensor readings were updated on Firebase without noticeable delay under normal Wi-Fi connectivity. The web dashboard refreshed automatically whenever new data was available. The buzzer successfully generated an alert whenever the AQI exceeded the predefined threshold.

4.6 Advantages

- Low-cost implementation.
- Real-time cloud monitoring.
- Wireless data transmission.
- Automatic emergency alert.
- Easy installation and maintenance.
- Suitable for smart homes and industries.

4.7 Limitations

- Requires continuous Wi-Fi connectivity.

- MQ-135 requires proper calibration.
- Accuracy depends on sensor quality.
- Limited battery backup without external power.

4.8 Discussion

The proposed system successfully demonstrated an IoT-based environmental monitoring solution capable of measuring air quality and temperature in real time. The integration of ESP32, Firebase, and a web dashboard provided an efficient and user-friendly monitoring platform. The system can be deployed in homes, laboratories, classrooms, offices, and industrial environments for continuous environmental monitoring.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

This project presented the design and implementation of an IoT-based Air Quality and Temperature Monitoring System using the ESP32 Development Board, MQ-135 Gas Sensor, and DS18B20 Temperature Sensor. The developed system continuously monitors environmental conditions and uploads the collected data to Firebase Realtime Database through a Wi-Fi network.

The web dashboard successfully displays the latest air quality, carbon monoxide concentration, temperature, and emergency status in real time. The active buzzer provides an immediate warning whenever the measured Air Quality Index exceeds the predefined threshold.

The experimental results demonstrate that the proposed system is reliable, low-cost, portable, and suitable for continuous environmental monitoring. The system can be effectively used in homes, educational institutions, laboratories, hospitals, offices, and industrial environments.

5.2 Future Scope

The proposed system can be further improved in several ways:

- Integration of additional environmental sensors such as humidity, smoke, LPG, and CO₂ sensors.
- Development of Android and iOS mobile applications.
- AI and Machine Learning based air quality prediction.
- SMS, Email, and WhatsApp notification during emergency conditions.
- GPS-based location tracking for outdoor monitoring.

- Solar-powered operation for remote locations.
- Data analytics and graphical trend visualization.
- Integration with Smart Home and Smart City platforms.

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